



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTTP/OPERATION/2023-24/OPN/2024/835251

Date:

18/01/2024(22:42:27)

Subject:REVISION:3 LMI ON PROCEDURE FOR HANDLING FIRE IN
TDBFP'S

LMI/SMPP/OPRN/001, Issue No: 1, Rev:03, Dated: Jan 2024 "PROCEDURE FOR HANDLING FIRE IN BFP'S" has been reviewed as per two years periodicity. No changes made in this revision.

LMI is attached and put up for further perusal and approval.

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On:18/01/2024(22:42:27) **Ref:**SMTTP/OPERATION/2023-24/OPN/2024/835251

Note No:2

Jist of LMI was presented by custodian during 02.02.2024 DPM. File is being forwarded for necessary approval please.

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Note No:5

Please review.

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Note No:6

Reviewed and no changes proposed.

Forwarded for further approval.

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Note No:7

No changes proposed. Forwarded for further approval.

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On:24/02/2024(18:40:45) **Ref:**SMTTP/OPERATION/2023-24/OPN/2024/835251

Note No:8

Reviewed and no changes proposed. Forwarded for further approval.

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Note No:9

kindly enclose OD/ OGN for review, please.

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Note No:10

This is Site LMI; OD/OGN is not available. The related IOM from CC-OS is attached at Annexure-1 for reference.

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Note No:11

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Note No:12

LMI & CC-OS IOM are reviewed and observed to be in order.

May be forwarded for approval.

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Note No:13

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Note No:14

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Note No:15

Please review page no1.

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Note No:16

Page no.1 has been modified and Attached as Annexure II & III.

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On:06/03/2024(19:24:07) **Ref:**SMTTP/OPERATION/2023-24/OPN/2024/835251

Note No:17

LMI forwarded for approval after review by all concerned sections

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Note No:18

Please discuss

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Note No:19

Annexure III may please be considered as final document for LMI revision.

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Note No:20

Reviewed, no changes proposed. Forwarded for further approval.

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Note No:21

Reviewed & no changes proposed

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Note No:22

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On:14/03/2024(18:52:45) **Ref:**SMTTP/OPERATION/2023-24/OPN/2024/835251

Note No:23

Forwarded for approval

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On:15/03/2024(07:34:44) **Ref:**SMTTP/OPERATION/2023-24/OPN/2024/835251

Note No:24

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On:15/03/2024(09:35:36) **Ref:**SMTTP/OPERATION/2023-24/OPN/2024/835251

LMI/SMPP/OPRN/001
Rev. No.: 03, Dt: Jan 2024

Local Management Instructions

Procedure for handling Fire in BFPs



Approved for implementation by HEAD OF PROJECT

Date

Prepared by OPERATION DEPARTMENT

NTPC Limited
Simhadri Thermal Power Project

Procedure for handling Fire in BFPs

LMI/SMPP/OPRN/001, Issue:01, Rev:3, Dt: Jan 2024

LMI (name): Procedure for handling fire in BFPs					
LMI no:: LMI/SMPP/OPRN/001					
Issue no	001	Revision no	03	Date	18.01.2024

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1.0 PURPOSE:-

This document is prepared to guide the operating personnel for taking necessary action in the event of fire in TDBFPs area including TG hall, TG 8.5 m & TG 0 mtr area.

2.0 SUPERSEDED DOCUMENTS & REVISION NOTES:

Superseded: 1. LMI/SMPP/OPRN/001, Issue No: 1, Rev:01, Dt: Jan 2020

2. LMI/SMPP/OPRN/001, Issue No: 1, Rev:02, Dt: Feb 2022

Revision Document: Rev:03, Dt: Jan 2024 - periodical review. No changes made in this revision.

3.0 SCOPE: -

This shall be applicable to all the 4 × 500 MW units of SMTTP-NTPC Ltd.

4.0 INTRODUCTION:

To cater to the feed water requirement of the boiler, three boiler feed water pumps have been provided. (Two TDBFPs & One MDBFP). During normal operation, both TDBFPs remain in service and MDBFP remains as standby on auto. TDBFP lubrication system comprises of Two no. Lub oil pumps, a JOP and an Emergency Oil Pump. In addition to the above, Lub oil is also used for governing system of TDBFPs. Lub oil tank for TDBFPs is located at turbine “0 m”, which feeds to all the lub oil pumps for lubrication & Governing system of BFPs. Despite taking adequate care, there may be chances of oil Leakage from the system at any elevation, potentially which can cause fire & can spread to the adjoining areas & can have catastrophic consequences.

In addition to it any fire prone debris like oil-soaked cotton waste or any insulation material or accumulated oil under the grills or trays can also lead to fire hazard.

5.0 OPERATING / HANDLING INSTRUCTIONS: -

5.1 SCHEDULED / ROUTINE CHECKS: -

- I. During annual overhauling Emulsifier operation to be checked for TDBFP lub oil system by recommissioning group in coordination with (TMD), (C&I), (Offsite) and (EMD). This actual operation is to be checked without covering auxiliaries and tank with plastic or tarpaulin cloth. During Fire protection sprinkler operation, sources of water entry into the system to be identified and attended. If any water entry is there, Fire protection sprinkler operation to be repeated till the leakage points are fully attended. Necessary protocols are to be maintained w.r.t above.

Resp.: Recom. (Team Leader)/AGM(TMD)/AGM(C&I)/AGM(EMD)

- II. Availability & Emulsifier deluge valve auto position status are to be checked in every shift.
- III. All hydrant points in TG area to be checked for availability of water.
- IV. Insulation of BFPs are intact & in order.
- V. Any oil accumulation in & around BFPs to be cleaned on urgent basis.

- VI. Housekeeping in BFP area to be ensured.
- VII. CCTVs also to be used for regular monitoring.
- VIII. Adequate alarms (Rate of lub oil tank level drop) for early detection of problem to be ensured.

Resp. (UCE)

5.2 SPECIFIC INSTRUCTIONS: -

THE FOLLOWING INSTRUCTIONS ARE TO BE FOLLOWED IN CASE FIRE IS OBSERVED IN TDBFP / TDBFP LOT AREA:

- i. Inform the SCE about the occurrence of the fire incident **Resp. (UCE/TG Desk Eng.)**
- ii. Inform Fire Station regarding fire in TDBFP & Other concerned HOD's **Resp. (SCE/UCE).**
- iii. Announce in the PA system about the fire. Stop all the works in progress in TG area & inform all the O&M personals to evacuate the TG area. **Resp. (UCE/TG Desk Eng.)**
- iv. Immediately trip the affected TDBFP & make sure that TDBFP barring gear valve does not open in order to bring the BFP in standstill condition at the earliest.
Resp. (UCE/TG Desk Eng.)
- v. Stop all LOPs of TDBFPs including JOP & EOP. if pumps are not stopping from MMI press the Local push Button (EPB). It may so happen that pumps status feedback in MMI is lost & not able to establish the status of pump. Please ensure that pumps have stopped from Local.
Resp. (UCE/TG Desk Eng.)
- vi. If fire has spread to TG 17 m area in the vicinity of Generator, Trip the unit & purge out H₂ from generator casing with CO₂ immediately.
Resp. (UCE/TG Desk Eng.)
- vii. Meanwhile rush to the LT switchgear & isolate the power supply of LOPs, EOP & JOP & ensure oil supply is completely isolated. Avoid going through affected unit TG area as visibility may be poor & will be unsafe. Use breathing apparatus while rushing to location.
Resp. (UCE/TG Desk Eng.)
- viii. If fire is in LOT area, Check the Emulsifier has operated on auto, if not, manually operate the emulsifier system.
Resp. (SCE/UCE)
- ix. Stop the vapor extraction Fans of LOT. **Resp. (UCE/TG Desk Eng.)**
- x. Stop the Air Washer Units blowers as it will supply air to fire. **Resp. (UCE/TG Desk Eng.)**
- xi. Fire to be quenched at the earliest from nearest hydrant point.
- xii. In case of any permit on fire hydrant system, alternate source of water like fire tender, sprinkler system to be used. In case of extreme emergency charge the fire hydrant system after taking consent from concerned maintenance. Dept.
- xiii. All efforts shall be put to extinguish the fire as early as possible.

Resp. SCE / AGM(TMD)/AC CISF (Fire wing)

- xiv. Comprehensively check if fire has spread to any other area in the TG hall. Extinguish the same with help of fire personnel.
- xv. Following areas to are to be checked
- xvi. TDBFP LOP area
- xvii. Control fluid room & DMCW pump area.
- xviii. CEPs & condenser subzero mtr area.
- xix. H2 rack & vac pump area
- xx. MOT room & oil canal.
- xxi. HPBP OSUs & main TG seal steam area.
- xxii. Main TG bearing area & adjoining insulations
- xxiii. All clear status of entire area to be paged in PA system while scanning the different areas of TG hall.
- xxiv. Depending on the point of leakage check if EOP can be kept in service **Resp. UCE**

5.3 POST FIRE CHECKS: -

- i. Identify the leakage /source of fire. **Resp.: AGM(TMD)/AGM(OPN)**
- ii. Affected area to be barricaded & post firefighting entire area to be cleaned. **Resp.: AGM (OPN)**
- iii. Restore the oil system as early as possible & start manual barring. **Resp.: AGM(O)/AGM(C&I)/AGM(TMD)/AGM(EMD)**
- iv. All power cables related to TDBFP are to be checked. **Resp.: AGM(EMD)**
- v. All control cables related to TDBFPs are to be checked. **Resp.: AGM(C&I)**
- vi. TDBFP & related pipelines are to be checked for integrity. **Resp.: AGM(TMD)**
- vii. If the turbine has to be brought to standstill without DC EOP and DC JOP, then the bearings are to be inspected. **Resp.: AGM(TMD)**
- viii. LOT Oil Sampling must be done to analyze oil for moisture content and other impurities. **Resp.: AGM(TMD)/AGM(O)/AGM (chem)**
- ix. All motors of oil system must be meggered to ensure adequate IR value. **Resp.: AGM (EMD)**
- x. C&I Switches and transmitters must be checked for water ingress and healthiness. **Resp.: AGM(C&I).**
- xi. Check all the parameters of BFPs in MMI **Resp.: AGM (OPN).**
- xii. All hangers & support structures are to be inspected. **Resp.: AGM (civil)/AGM(TMD)**
- xiii. Insulation of the affected area & adjoining area to be removed & fresh insulation & cladding to be done. **Resp.: AGM(TMD)**

5.4 SYSTEM RESTORATION

- i. After assessment of damage, PTWs to be issued to respective departments for restoration & repair works. **Resp. UCE**
- ii. Clearance to be taken after completion of work from all departments before charging the system. **Resp. UCE**
- iii. Ensure all old insulation is removed & new insulation is done with fire resistant material. **Resp. AGM(O) / AGM(TMD)**
- iv. Area should be cleaned before charging the system. & proper illumination to be ensured. **Resp. UCE**
- v. After keeping LOPs in service, check for leakages at all elevations. **Resp. UCE**
- vi. All protections & interlocks of affected BFP to be checked with minimum simulations. **Resp. UCE**
- vii. Operation of all the valve & actuators of the system to be checked from local & remote with proper feedback & the same to be recorded. **Resp. UCE/ TG desk Engr**
- viii. Governing characteristics to be checked & deviation from previous value, if any to be recorded. **Resp. UCE**
- ix. All transmitters including vibration, temp & other parameters are to be made available. In absence of any critical parameter, BFP should not be rolled. **Resp. UCE/ C&I**
- x. After completion of all checks & clearances BFP to be rolled & all the parameter to be recorded in logbook. **Resp. UCE/ TG desk Engr**
- xi. BFP to be soaked adequately before loading the same. During soaking period all elevations of TG should be checked. Unidentified area or uncleaned area in the system or vicinity may again catch fire as the system heats up. **Resp. UCE/ TG desk Engr**
- xii. CISF fire wing to be deployed at TG 0 mtr, 8.5 m & 17 m from the point of Lub. Oil system charging to 48 hrs. Post TDBFP loading. **Resp. SCE**
- xiii. After ensuring all the parameters & local clearance BFP to be loaded **Resp. UCE/ TG desk Engr**
- xiv. All parameters of BFP to be recorded in log sheets after loading the same. **Resp. UCE/ TG desk Engr**

6.0 Record: All the incidences of BFP fire & actions taken with respect to the given LMI are to be recorded. Deviations if any are to be noted along with reasons.

7.0 Review: This document is to be reviewed once in two years by HOD (O).

END